

```

}
prob=gam.gammq(0.5*df,0.5*chisq);
minij = nni < nnj ? nni-1 : nnj-1;
cramrv=sqrt(chisq/(sum*minij));
ccc=sqrt(chisq/(chisq+sum));
}

```

Chi-square probability function.

CITED REFERENCES AND FURTHER READING:

- Agresti, A. 2002, *Categorical Data Analysis*, 2nd ed. (New York: Wiley).
- Mickey, R.M., Dunn, O.J., and Clark, V.A. 2004, *Applied Statistics: Analysis of Variance and Regression*, 3rd ed. (New York: Wiley).
- Norusis, M.J. 2006, *SPSS 14.0 Guide to Data Analysis* (Englewood Cliffs, NJ: Prentice-Hall).

14.5 Linear Correlation

We next turn to measures of association between variables that are ordinal or continuous, rather than nominal. Most widely used is the *linear correlation coefficient*. For pairs of quantities (x_i, y_i) , $i = 0, \dots, N - 1$, the linear correlation coefficient r (also called the product-moment correlation coefficient, or *Pearson's r*) is given by the formula

$$r = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2} \sqrt{\sum_i (y_i - \bar{y})^2}} \quad (14.5.1)$$

where, as usual, $\bar{x}$ is the mean of the x_i 's and $\bar{y}$ is the mean of the y_i 's.

The value of r lies between -1 and 1 , inclusive. It takes on a value of 1 , termed “complete positive correlation,” when the data points lie on a perfect straight line with positive slope, with x and y increasing together. The value 1 holds independent of the magnitude of the slope. If the data points lie on a perfect straight line with negative slope, y decreasing as x increases, then r has the value -1 ; this is called “complete negative correlation.” A value of r near zero indicates that the variables x and y are *uncorrelated*.

When a correlation is known to be significant, r is one conventional way of summarizing its strength. In fact, the value of r can be translated into a statement about what residuals (root-mean-square deviations) are to be expected if the data are fitted to a straight line by the least-squares method (see §15.2, especially equation 15.2.13). Unfortunately, r is a rather poor statistic for deciding *whether* an observed correlation is statistically significant and/or whether one observed correlation is significantly stronger than another. The reason is that r is ignorant of the individual distributions of x and y , so there is no universal way to compute its distribution in the case of the null hypothesis.

About the only general statement that can be made is this: If the null hypothesis is that x and y are uncorrelated, and if the distributions for x and y each have enough convergent moments (“tails” die off sufficiently rapidly), and if N is large (typically

> 500), then r is distributed approximately normally, with a mean of zero and a standard deviation of $1/\sqrt{N}$. In that case, the (double-sided) significance of the correlation, that is, the probability that $|r|$ should be larger than its observed value in the null hypothesis, is

$$\operatorname{erfc}\left(\frac{|r|\sqrt{N}}{\sqrt{2}}\right) \quad (14.5.2)$$

where $\operatorname{erfc}(x)$ is the complementary error function, equation (6.2.10), computed by the routines `Erf.erfc` or `erfcc` of §6.2. A small value of (14.5.2) indicates that the two distributions are significantly correlated. (See expression 14.5.9 below for a more accurate test.)

Most statistics books try to go beyond (14.5.2) and give additional statistical tests that can be made using r . In almost all cases, however, these tests are valid only for a very special class of hypotheses, namely that the distributions of x and y jointly form a *binormal* or *two-dimensional Gaussian* distribution around their mean values, with joint probability density

$$p(x, y) dx dy = \text{const.} \times \exp\left[-\frac{1}{2}(a_{00}x^2 - 2a_{01}xy + a_{11}y^2)\right] dx dy \quad (14.5.3)$$

where a_{00} , a_{01} , and a_{11} are arbitrary constants. For this distribution r has the value

$$r = -\frac{a_{01}}{\sqrt{a_{00}a_{11}}} \quad (14.5.4)$$

There are occasions when (14.5.3) may be known to be a good model of the data. There may be other occasions when we are willing to take (14.5.3) as at least a rough-and-ready guess, since many two-dimensional distributions do resemble a binormal distribution, (that is, a two-dimensional Gaussian) at least not too far out on their tails. In either situation, we can use (14.5.3) to go beyond (14.5.2) in any of several directions:

First, we can allow for the possibility that the number N of data points is not large. Here, it turns out that the statistic

$$t = r \sqrt{\frac{N-2}{1-r^2}} \quad (14.5.5)$$

is distributed in the null case (of no correlation) like Student's t -distribution with $\nu = N - 2$ degrees of freedom, whose two-sided significance level is given by $1 - A(t|\nu)$ (equation 6.14.11) [1]. As N becomes large, this significance and (14.5.2) become asymptotically the same, so that one never does worse by using (14.5.5), even if the binormal assumption is not well substantiated.

Second, when N is only moderately large (≥ 10), we can compare whether the difference of two significantly nonzero r 's, e.g., from different experiments, is itself significant. In other words, we can quantify whether a change in some control variable significantly alters an existing correlation between two other variables. This is done by using *Fisher's z-transformation* to associate each measured r with a corresponding z :

$$z = \frac{1}{2} \ln\left(\frac{1+r}{1-r}\right) \quad (14.5.6)$$

Then, each z is approximately normally distributed with a mean value

$$\bar{z} = \frac{1}{2} \left[\ln \left(\frac{1 + r_{\text{true}}}{1 - r_{\text{true}}} \right) + \frac{r_{\text{true}}}{N - 1} \right] \quad (14.5.7)$$

where r_{true} is the actual or population value of the correlation coefficient, and with a standard deviation

$$\sigma(z) \approx \frac{1}{\sqrt{N - 3}} \quad (14.5.8)$$

Equations (14.5.7) and (14.5.8), when they are valid, give several useful statistical tests [1]. For example, the significance level at which a measured value of r differs from some hypothesized value r_{true} is given by

$$\text{erfc} \left(\frac{|z - \bar{z}| \sqrt{N - 3}}{\sqrt{2}} \right) \quad (14.5.9)$$

where z and $\bar{z}$ are given by (14.5.6) and (14.5.7), with small values of (14.5.9) indicating a significant difference. (Setting $\bar{z} = 0$ makes expression 14.5.9 a more accurate replacement for expression 14.5.2 above.) Similarly, the significance of a difference between two measured correlation coefficients r_1 and r_2 is

$$\text{erfc} \left(\frac{|z_1 - z_2|}{\sqrt{2} \sqrt{\frac{1}{N_1 - 3} + \frac{1}{N_2 - 3}}} \right) \quad (14.5.10)$$

where z_1 and z_2 are obtained from r_1 and r_2 using (14.5.6), and where N_1 and N_2 are, respectively, the number of data points in the measurement of r_1 and r_2 .

All of the significances above are two-sided. If you wish to disprove the null hypothesis in favor of a one-sided hypothesis, such as that $r_1 > r_2$ (where the sense of the inequality was decided *a priori*), then (i) if your measured r_1 and r_2 have the *wrong* sense, you have failed to demonstrate your one-sided hypothesis, but (ii) if they have the right ordering, you can multiply the significances given above by 0.5, which makes them more significant.

But keep in mind: These interpretations of the r statistic can be completely meaningless if the joint probability distribution of your variables x and y is too different from a binormal distribution.

`void pearsn(VecDoub_I &x, VecDoub_I &y, Doub &r, Doub &prob, Doub &z)`

[stattests.h](#)

Given two arrays $x[0..n-1]$ and $y[0..n-1]$, this routine computes their correlation coefficient r (returned as r), the p -value at which the null hypothesis of zero correlation is disproved (prob whose small value indicates a significant correlation), and Fisher's z (returned as z), whose value can be used in further statistical tests as described above.

```
{
    const Doub TINY=1.0e-20;
    Beta beta;
    Int j,n=x.size();
    Doub yt,xt,t,df;
    Doub syy=0.0,sxy=0.0,sxx=0.0,ay=0.0,ax=0.0;
    for (j=0;j<n;j++) {
        ax += x[j];
        ay += y[j];
    }
```

Will regularize the unusual case of complete correlation.

Find the means.